Early-Phase Clinical Pharmacology: Biostatistics Deliverables Pipeline

Protocol review → CSR draft · every document, dataset, and statistical output, with its owner, standard, and QC

> ***Reference map, not a track record.*** *This register maps the deliverables of a Phase 1 clinical pharmacology study and how each would be produced and checked. It is not a list of deliverables the author produced (the home page labels what he led or supported). A short appendix sets out where AI tools could assist under controls; it is a proposal, not an adopted policy.*
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> ***Scope & how to read this.*** *132 deliverables across five stages of a Phase 1 clinical-pharmacology study (e.g., First-in-Human SAD/MAD with PK/PD, NCA + popPK, safety), each with its owner, inputs, outputs, governing standard, and QC method. The at-a-glance tables are below; the* ***full register with every field is the companion `.xlsx`****, and the* ***one-page visual is the `.pdf` map****. Standards as of October 2026: CDISC SDTMIG v3.4, ADaMIG v1.3, and Define-XML v2.1 (all FDA-required); ICH E3, E6(R3), E9(R1), and M11; systems designed to meet 21 CFR Part 11; ISPE GAMP 5.*

How the pipeline flows

Five sequential stages, each closed by a gate. A **CDISC data-standards track** (SDTM → ADaM → Define-XML + reviewer guides) runs through Stages 3 to 5, and a **quality track** (independent double programming, engine-of-record records, audit trail, named sign-off) runs across all five.

Stage	Focus	Gate
S1: Study Design & Protocol	Biostatistics shapes the design: estimands, endpoints, sample-size rationale, randomization, and the statistical sections of the protocol.	Protocol approved / finalized
S2: Planning & Specifications	Everything is specified before data: the SAP, TLF shells, ADaM/SDTM specs, the randomization schedule, the data-review and QC plans.	SAP & specifications final
S3: Data & CDISC	Raw data becomes standardized, conformant, traceable CDISC datasets (SDTM → ADaM) with reviewer guides and Define-XML, under full data integrity controls.	Database lock
S4: Analysis & QC	TLFs and PK/PD analyses are produced and independently double-programmed; reported numbers come from validated programs and engines (Phoenix WinNonlin or PKNCA for NCA).	All outputs QC-passed
S5: Reporting & CSR	Results become the CSR: statistical methods and results text, in-text tables/figures, appendices, the submission package, and cross-document consistency.	CSR draft delivered

S1: Study Design & Protocol

Biostatistics shapes the design: estimands, endpoints, sample-size rationale, randomization, and the statistical sections of the protocol.

ID	Deliverable	Owner	Standard	QC
S1-01	**Biostatistics input to protocol: design rationale & study design selection**	Lead biostatistician	ICH M11 (CeSHarP protocol template) design section; ICH E9 (design principles, bias control); ICH E6(R3) GCP; FDA/EMA FIH guidance	Independent biostatistician peer review; cross-check design vs objectives/estimands; medical/clin-pharm/regulatory review; protocol team consensus
S1-02	**Objectives → estimands → endpoints alignment (ICH E9(R1) estimand framework)**	Lead biostatistician	ICH E9(R1) estimand addendum; ICH E9; ICH M11 (estimand placement in protocol); ICH E3	Independent biostatistician review of estimand completeness (all 5 attributes, every ICE addressed); clinical & clin-pharm alignment review; consistency check objectives ↔ estimands ↔ endpoints ↔ analysis sets
S1-03	**Statistical considerations / analysis-overview section of protocol**	Lead biostatistician	ICH E9 / E9(R1); ICH M11 statistical-considerations section; ICH E3 (consistency with future CSR methods); ICH E6(R3)	Peer review by independent biostatistician; alignment check vs estimands and vs planned SAP scope; cross-functional protocol review
S1-04	**Analysis populations / analysis-set definitions (protocol-level)**	Lead biostatistician	ICH E9 / E9(R1); ICH M11; CDISC ADaMIG v1.3 (population-flag conventions, e.g., SAFFL/PKFL); ICH E3	Independent review of set definitions for mutual consistency and estimand linkage; clin-pharm review of PK-evaluable exclusion rules; traceability check to ADSL flag plan
S1-05	**Sample-size / power justification (or documented rationale for no formal sizing in FIH)**	Lead biostatistician	ICH E9 (sample size & assumptions); ICH M11; FDA BA/BE & food-effect guidance; EMA bioequivalence/early-phase guidance; ICH E9(R1) (sizing tied to estimand)	Independent re-derivation/double-check of sample-size or precision calculation (independent code/program run); review of assumptions vs source data; documentation of method and seed/software version
S1-06	**Randomization design & specification (incl. sentinel dosing & active:placebo allocation)**	Lead biostatistician	ICH E9 (randomization & blinding); ICH E6(R3) GCP; ICH M11; 21 CFR Part 11 (controlled, audit-trailed list generation); sponsor randomization SOP	Independent biostatistician review of design spec; verification that block/ratio/sentinel logic enforces intended allocation; segregation-of-duties check (spec author ≠ schedule generator); blinding-integrity review
S1-07	**DSMB/SRC/SEC charter: statistical decision rules & dose-**	Lead biostatistician	ICH E9 (data monitoring, decision rules); ICH E6(R3) GCP;	Independent review of decision-rule logic for unambiguity and safety

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	escalation criteria		FDA guidance on Data Monitoring Committees; FIH guidance (escalation/stopping); ICH M11	conservatism; consistency check vs protocol dose levels and stopping rules; committee/regulatory review; (if model-based) operating-characteristics simulation review
S1-08	**Dose-escalation operating-characteristics simulation (design-stage)**	Lead biostatistician	ICH E9 / E9(R1); FDA adaptive-design & FIH guidance; reproducibility per 21 CFR Part 11 (versioned, seeded code); sponsor simulation SOP	Independent re-run/replication of simulation with documented seed & environment; code review; scenario-grid completeness review; sanity-check of OC against design intuition
S1-09	**Interim / adaptive-design elements specification (if any)**	Lead biostatistician	ICH E9 / E9(R1); FDA Adaptive Designs for Clinical Trials guidance; EMA reflection paper on adaptive designs; ICH M11; 21 CFR Part 11	Independent statistical review of adaptation rules & error-rate control; simulation of operating characteristics; firewall/independence review; documentation of 'N/A' decision where no adaptation
S1-10	**PK precision / variability assumptions memo (CV%, BLQ/LLOQ & sampling-scheme adequacy)**	Lead biostatistician	ICH E9; FDA/EMA PK and BA/BE guidance; sponsor NCA/PK SOP; reproducible assumptions	Joint biostatistician–PK scientist review; cross-check sampling adequacy vs predicted t1/2; sensitivity of precision to CV% assumptions; documentation of assumption sources
S1-11	**Biostatistician protocol review & sign-off**	Lead biostatistician	ICH E9 / E9(R1); ICH M11; ICH E6(R3) GCP; 21 CFR Part 11 (electronic signature & audit trail); sponsor protocol-approval SOP	Comment-tracking & resolution log; independent (second-biostatistician) consistency check on objectives ↔ estimands ↔ endpoints; signed/audited approval; amendment version control
S1-12	**Emergency unblinding / code-break plan (design-stage definition)**	Lead biostatistician	ICH E6(R3) GCP; ICH E9 (blinding integrity); 21 CFR Part 11 (audit trail of unblinding); pharmacovigilance regs; sponsor unblinding SOP	Independent review of firewall & segregation-of-duties; verification unblinding pathway is auditable and minimally disruptive to the blind; cross-check vs DSMB charter and randomization spec
S1-13	**Blinded Sample-Size Re-Estimation (BSSR) Plan: design-stage provision**	Lead biostatistician	ICH E9 / E9(R1); ICH E6(R3) GCP	Biostatistician review and sign-off; no unblinded data touched

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S1-14	**Estimand-to-design traceability & assumptions log (intercurrent-event strategy register)**	Lead biostatistician	ICH E9(R1) estimand framework	Biostatistician + medical reviewer sign-off

Stage gate: Protocol approved / finalized. 14 deliverables.

S2: Planning & Specifications

Everything is specified before data: the SAP, TLF shells, ADaM/SDTM specs, the randomization schedule, the data-review and QC plans.

ID	Deliverable	Owner	Standard	QC
S2-01	**Statistical Analysis Plan (SAP)**	Lead biostatistician	ICH E9 + E9(R1) estimands, ICH E3 (alignment with CSR), ICH E6(R3) GCP, ICH M11; version-controlled, dated, signed	Senior biostatistician peer review; cross-check vs protocol & estimands; documented sign-off; formal version control of amendments
S2-02	**SAP Amendment & Version-Control Log**	Lead biostatistician	ICH E9, ICH E3, ICH E6(R3), 21 CFR Part 11 (audit trail), ALCOA++	Independent review of each change; timestamp vs lock/unblinding verified; Part 11 audit trail retained
S2-03	**Analysis-Set / Population Definitions**	Lead biostatistician	ADaMIG v1.3 (population flags), ICH E9(R1), CDISC CT	Independent reviewer confirms flag logic vs SAP; double-programmed in ADSL; reconciled at blinded data review
S2-04	**TLF Shells / Mock-Up Shells (Tables, Listings, Figures)**	Lead biostatistician	ICH E3 (in-text/appendix structure), sponsor output standards, CDISC CT for terminology	Peer/medical-writer/ clinical review of shells; titles/footnotes vs SAP; dummy-data render test for layout
S2-05	**TLF Inventory / Table of Contents**	Lead biostatistician	ICH E3, sponsor standards	Cross-check vs SAP analyses (no orphan/missing outputs); peer review
S2-06	**ADaM Dataset Specifications (ADSL)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1, traceability to SDTM	Spec peer review; downstream independent double-programming of ADSL; Pinnacle 21 ADaM conformance
S2-07	**ADaM Dataset Specification (ADPC: PK Concentrations)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1, ICH PK reporting practice	Spec review vs PK plan (BLQ/time rules); independent double-programming of ADPC; reconciliation to SDTM PC
S2-08	**ADaM Dataset Specification (ADPP:**	Stat programmer	ADaMIG v1.3, CDISC CT (PK parameters),	Spec review vs parameter list; ADPP

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	PK Parameters)		Define-XML v2.1	reconciled to validated Phoenix output; independent double-programming of structure
S2-09	**ADaM Dataset Specification (ADAE: Adverse Events)**	Stat programmer	ADaMIG v1.3 (OCCDS), MedDRA, CDISC CT, Define-XML v2.1	Spec review of TEAE logic; independent double-programming; reconcile vs SDTM AE
S2-10	**ADaM Dataset Specifications (ADLB / ADVS / ADEG: Labs, Vitals, ECG)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1	Spec review of baseline/window/unit logic; independent double-programming; SDTM reconciliation
S2-11	**ADaM Dataset Specification (ADPD: Pharmacodynamics)**	Stat programmer	ADaMIG v1.3, CDISC CT, Define-XML v2.1	Spec review vs PD plan; independent double-programming; SDTM reconciliation
S2-12	**SDTM Mapping Specification**	Stat programmer	CDISC SDTMIG v3.4, Controlled Terminology, Define-XML v2.1	Independent review vs SDTMIG/CT; Pinnacle 21 rule mapping; traceability check raw->SDTM
S2-13	**Annotated CRF (aCRF): Preparation & Review**	Stat programmer	CDISC SDTMIG v3.4, Define-XML v2.1, CDISC aCRF conventions	Page-by-page review vs mapping spec; consistency with Define-XML origin; peer review
S2-14	**Programming Specifications / Derivation Specs**	Stat programmer	ADaMIG v1.3, Define-XML v2.1 (value-level), CDISC CT	Independent spec review; basis for double-programming reconciliation; traceability to SAP
S2-15	**PK Analysis Plan (NCA detail)**	PK scientist	ICH PK reporting practice, regulatory NCA guidance, Phoenix WinNonlin (validated NCA engine)	PK scientist + biostatistician review; rules cross-checked vs SAP; Phoenix template qualified
S2-16	**popPK / Exposure-Response Modeling Analysis Plan (PMAP)**	Pharmacometrician	Regulatory popPK/E-R guidance, ICH E9(R1), software validation; NONMEM/Monolix as analysis engines	Independent pharmacometrics review; model-qualification criteria pre-specified; run records reproducible
S2-17	**NONMEM/Monolix Analysis Dataset Specification (PK/PD modeling dataset)**	Pharmacometrician	popPK dataset conventions, traceability to ADaM, software validation	Independent check vs ADaM source; structure/covariate completeness verified; reproducibility
S2-18	**Randomization Schedule Generation**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11 (audit trail, access control), ALCOA++	Independent QC of allocation vs spec (ratios/blocks/strata); seed/reproducibility verified; secured storage
S2-19	**Randomization Specification**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11	Independent review vs protocol; IRT UAT alignment;

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				documented approval
S2-20	**Unblinding / Code-Break Plan**	Lead biostatistician	ICH E6(R3), ICH E9, 21 CFR Part 11 (audit trail), ALCOA++	SOP/charter alignment review; role/firewall verification; emergency-unblinding log tested
S2-21	**Data Review Plan / Blinded Data-Review Plan**	Lead biostatistician	ICH E9 (pre-unblinding decisions), ICH E6(R3), ALCOA++	Peer review; decisions recorded with timestamp before unblinding; consistency with SAP
S2-22	**Define-XML Planning / Metadata Strategy**	Stat programmer	Define-XML v2.1, CDISC SDTMIG v3.4 / ADaMIG v1.3, Controlled Terminology	Metadata completeness vs specs; Pinnacle 21 define checks; value-level metadata review
S2-23	**Controlled Terminology Plan / CT Version Pinning**	Stat programmer	CDISC Controlled Terminology, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1	Version-pin verification; Pinnacle 21 CT checks; extensible-codelist review
S2-24	**Validation / QC Plan (Independent Double-Programming Strategy)**	Lead biostatistician	ICH E6(R3), 21 CFR Part 11, ALCOA++, CDISC conformance; sponsor validation SOPs	Risk-based assignment reviewed/approved; double-programming mandatory for key outputs; audit-ready
S2-25	**Pseudo / Dummy Data for Shell Testing**	Stat programmer	CDISC structures (for realism), sponsor standards; non-PHI by construction	Structure conformance to ADaM specs; coverage of edge cases (BLQ, missing, windows)
S2-26	**Data-Transfer Specifications (DTS / interface review)**	Data mgmt	CDISC SDTMIG v3.4 (target), 21 CFR Part 11, ALCOA++, sponsor DM SOPs	Test-transfer review; key/format/unit verification; reconciliation procedure agreed
S2-27	**Dose-Escalation / Operating-Characteristics Simulation Plan & Outputs**	Lead biostatistician	ICH E9 + E9(R1), reproducibility/software validation	Independent review of assumptions/code; seed/reproducibility documented; results sanity-checked
S2-28	**Interim Analysis Plan / DSMB statistical output specifications & firewall (SAP-level)**	Independent unblinded statistician + lead biostatistician	ICH E9; ICH E6(R3); FDA/EMA DSMB guidance	Independent unblinded statistician; firewall enforced
S2-29	**Bioanalytical data-transfer & PK sample reconciliation specification (LC-MS/MS lab interface)**	PK scientist + biostat programming + bioanalytical lab	CDISC SDTMIG PC domain; FDA Bioanalytical Method Validation guidance; ALCOA++	Double-check vs lab dictionary; PK scientist sign-off
S2-30	**Value-Level Metadata (VLM) specification for Define-XML**	Biostat programming / metadata lead	Define-XML v2.1; CDISC ADaMIG v1.3 / SDTMIG v3.4	Independent review vs datasets; Pinnacle 21 Define validation
S2-31	**Software / computing environment validation &**	Biometrics systems / QA + statistical programming	21 CFR Part 11; GAMP 5; ICH E6(R3)	QA review; reported numbers always from validated engines

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	installation qualification (IQ/OQ) records			

Stage gate: SAP & specifications final. 31 deliverables.

S3: Data & CDISC

Raw data becomes standardized, conformant, traceable CDISC datasets (SDTM → ADaM) with reviewer guides and Define-XML, under full data integrity controls.

ID	Deliverable	Owner	Standard	QC
S3-01	**EDC / raw data extracts & Data Transfer Agreement (DTA) interface**	Data mgmt (biostat/stat programmer co-reviewer)	21 CFR Part 11, ALCOA++, ICH E6(R3); CDISC CT for code lists; sponsor DTA SOP	Transfer-spec conformance check on each load; record counts & checksum reconciliation; test-transfer (dry-run) before first production transfer
S3-02	**Biostat review of edit checks / data queries / data-review listings**	Lead biostatistician / Stat programmer (with Data mgmt)	ICH E6(R3) GCP, 21 CFR Part 11, ALCOA++; Data Review Plan	Independent review of listing logic; query resolution tracked to closure; reconciliation against SAP-defined critical variables
S3-03	**Annotated CRF (aCRF)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4, Define-XML v2.1, Controlled Terminology; CDISC aCRF conventions	Peer review against final CRF and SDTM specs; Pinnacle 21 / Define linkage check that every annotated origin resolves
S3-04	**SDTM datasets: DM (Demographics)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4, Controlled Terminology, Define-XML v2.1	Independent double-programming + Pinnacle 21 conformance; reconciliation to raw and randomization
S3-05	**SDTM datasets: PC (Pharmacokinetic Concentrations)**	Stat programmer (CDISC/SDTM lead) with PK scientist	CDISC SDTMIG v3.4 (Findings), Define-XML v2.1, value-level metadata; CT for units/analytes	Independent double-programming + Pinnacle 21; 100% reconciliation to bioanalytical transfer (record count, value, BLQ flags); nominal-vs-actual time check
S3-06	**SDTM datasets: PP (Pharmacokinetic Parameters)**	Stat programmer (CDISC/SDTM lead) with PK scientist	CDISC SDTMIG v3.4 (PP domain), CT for PK parameters (PKPARMCD), Define-XML v2.1, RELREC linkage to PC	Reconciliation to Phoenix output (the engine of record); independent double-programming of the SDTM structure; Pinnacle 21
S3-07	**SDTM datasets: EX (Exposure)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Interventions), Define-XML v2.1, Controlled Terminology	Independent double-programming + Pinnacle 21; reconciliation to randomization and to

ID	Deliverable	Owner	Standard	QC
				PC time-after-dose derivation
S3-08	**SDTM datasets: AE (Adverse Events)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (MedDRA coding)	CDISC SDTMIG v3.4 (Events), MedDRA (versioned), Define-XML v2.1, CT	Independent double-programming + Pinnacle 21; MedDRA coding QC; reconciliation to safety database/SAE recon
S3-09	**SDTM datasets: LB (Laboratory)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Findings), CT/LOINC, Define-XML v2.1 value-level metadata, unit standardization	Independent double-programming + Pinnacle 21; 100% reconciliation to lab transfer; unit conversion QC
S3-10	**SDTM datasets: VS (Vital Signs)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to raw
S3-11	**SDTM datasets: EG (ECG)**	Stat programmer (CDISC/SDTM lead) with PK/cardiac-safety scientist	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1; time-matched to PC for C-QTc	Independent double-programming + Pinnacle 21; reconciliation to ECG core-lab transfer; PK time-matching QC
S3-12	**SDTM datasets: DS (Disposition)**	Stat programmer (CDISC/SDTM lead)	CDISC SDTMIG v3.4 (Events), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to enrollment/randomization counts
S3-13	**SDTM datasets: MH (Medical History)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (coding)	CDISC SDTMIG v3.4 (Events), MedDRA, CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-14	**SDTM datasets: CM (Concomitant/Prior Medications)**	Stat programmer (CDISC/SDTM lead) with Data mgmt (WHODrug coding)	CDISC SDTMIG v3.4 (Interventions), WHODrug (versioned), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-15	**SDTM datasets: PD/DV and other supporting domains (PD/PE/SU/QS, DV, CO, SE/TA/TV/TE trial design, SUPPQUAL, RELREC)**	Stat programmer (CDISC/SDTM lead) with PD/PK scientists	CDISC SDTMIG v3.4 (Findings/Events/Special-Purpose/Trial-Design/SUPP/RELREC), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; RELREC integrity check (PC<->PP); deviation-classification reconciliation
S3-16	**SDTM Define-XML (define.xml + stylesheet)**	Stat programmer (CDISC/SDTM lead)	CDISC Define-XML v2.1, SDTMIG v3.4, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; peer review; origin/aCRF cross-link check
S3-17	**SDTM Reviewer Guide (cSDRG)**	Stat programmer (CDISC/SDTM lead) / Lead biostatistician	PHUSE cSDRG template, CDISC SDTMIG v3.4, Define-XML v2.1	Peer review; consistency with Define-XML and Pinnacle 21 explanations; sign-off
S3-18	**ADaM datasets: ADSL (participant-level**	Stat programmer (ADaM lead) / Lead	CDISC ADaMIG v1.3, Define-XML v2.1,	Independent double-programming +

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	analysis dataset)	biostatistician	traceability to SDTM, CT	Pinnacle 21 ADaM checks; population-flag reconciliation to disposition/randomization
S3-19	**ADaM datasets: ADPC (PK Concentrations Analysis)**	Stat programmer (ADaM lead) with PK scientist	CDISC ADaMIG v1.3 (BDS / PK ADaM), Define-XML v2.1, traceability to PC/EX, CT	Independent double-programming + Pinnacle 21; time-after-dose & BLQ-rule QC; reconciliation to PC
S3-20	**ADaM datasets: ADPP (PK Parameters Analysis)**	Stat programmer (ADaM lead) with PK scientist	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to PP/Phoenix, CT (PK params)	Reconciliation to Phoenix (engine of record); independent double-programming; Pinnacle 21
S3-21	**ADaM datasets: ADAE (Adverse Events Analysis)**	Stat programmer (ADaM lead) / Lead biostatistician	CDISC ADaMIG v1.3 OCCDS, MedDRA, Define-XML v2.1, traceability to AE, CT	Independent double-programming + Pinnacle 21; TRTEMFL logic QC; reconciliation to SDTM AE & SAE database
S3-22	**ADaM datasets: ADLB (Laboratory Analysis)**	Stat programmer (ADaM lead)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1 value-level metadata, traceability to LB, CT	Independent double-programming + Pinnacle 21; baseline/change & category-derivation QC; reconciliation to LB
S3-23	**ADaM datasets: ADVS (Vital Signs Analysis)**	Stat programmer (ADaM lead)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to VS, CT	Independent double-programming + Pinnacle 21; baseline/change QC; reconciliation to VS
S3-24	**ADaM datasets: ADEG (ECG Analysis)**	Stat programmer (ADaM lead) with cardiac-safety/PK scientist	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to EG/PC, CT; ICH E14/concentration-QTc practice	Independent double-programming + Pinnacle 21; QTc category & time-matching QC; reconciliation to EG
S3-25	**ADaM datasets: ADPD and other analysis datasets (ADPD, ADCM, ADMH, deviation/exposure analysis as applicable)**	Stat programmer (ADaM lead) with PD/PK scientists	CDISC ADaMIG v1.3 BDS/OCCDS, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21; PK-PD linkage & baseline/change QC
S3-26	**ADaM Define-XML (define.xml + stylesheet)**	Stat programmer (ADaM lead)	CDISC Define-XML v2.1, ADaMIG v1.3, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; derivation-method review; traceability check
S3-27	**ADaM Reviewer Guide (ADRG)**	Stat programmer (ADaM lead) / Lead biostatistician	PHUSE ADRG template, CDISC ADaMIG v1.3, Define-XML v2.1	Peer review; consistency with Define-XML, SAP and Pinnacle 21 explanations; sign-off
S3-28	**CDISC conformance / validation (Pinnacle 21) report: SDTM & ADaM**	Stat programmer (CDISC lead) / Lead biostatistician	CDISC conformance rules, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, FDA/PMDA	Validated Pinnacle 21 engine; documented triage/disposition of every finding; reviewer-

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			validator rules	guide reconciliation
S3-29	**PK bioanalytical sample reconciliation (concentration data load & BLQ handling)**	PK scientist / Stat programmer (with bioanalytical lab & Data mgmt)	FDA/ICH M10 bioanalytical method validation, 21 CFR Part 11, ALCOA++; PK analysis plan BLQ rules	100% sample-accountability reconciliation; independent check of BLQ/LLOQ flagging; record-count/value match to transfer
S3-30	**Blinded data review outputs (data-review listings & blinded review meeting package)**	Lead biostatistician (with Data mgmt, Clinical, PK)	ICH E9/E9(R1), ICH E6(R3); Data Review Plan; 21 CFR Part 11	Independent review of listings; documented blinded decisions (minutes) prior to unblinding; reconciliation to SAP rules
S3-31	**Dry-run / test datasets & dry-run TLF execution**	Stat programmer / Lead biostatistician	Sponsor validation/QC plan; CDISC SDTMIG v3.4/ADaMIG v1.3; 21 CFR Part 11 (program version control)	Independent double-programming on dry-run; shell-vs-output reconciliation; defect tracking to closure
S3-32	**Data cleaning / soft lock**	Data mgmt (with Lead biostatistician sign-off)	ICH E6(R3), 21 CFR Part 11, ALCOA++; sponsor lock SOP	Lock checklist completion; cross-functional reconciliation sign-off; audit-trail review
S3-33	**Database hard lock**	Data mgmt (cross-functional sign-off incl. Lead biostatistician)	ICH E6(R3), 21 CFR Part 11 (audit trail/e-records), ALCOA++; sponsor lock SOP	Formal hard-lock checklist & multi-party sign-off; audit-trail/lock verification; snapshot integrity check
S3-34	**Traceability (raw -> SDTM -> ADaM) matrix**	Stat programmer (CDISC lead) / Lead biostatistician	CDISC SDTMIG v3.4/ADaMIG v1.3/Define-XML v2.1 traceability principles; ICH E6(R3), ALCOA++	Cross-link verification raw<->SDTM<->ADaM; peer review; consistency with define/reviewer guides
S3-35	**Controlled Terminology & coding-dictionary version management (CDISC CT, MedDRA, WHODrug)**	Data mgmt / Stat programmer (CDISC lead)	CDISC Controlled Terminology (versioned), MedDRA, WHODrug; Define-XML v2.1; 21 CFR Part 11	Version-lock verification; CT-conformance via Pinnacle 21; coding QC against locked dictionaries
S3-36	**Randomization QC / verification report (schedule vs IXRS/EDC reconciliation)**	Independent statistician (not the schedule generator)	ICH E6(R3); 21 CFR Part 11; ALCOA++	Independent double-check; segregation of duties from generator
S3-37	**Emergency unblinding / code-break event log & reconciliation**	Unblinded pharmacovigilance / IXRS admin + biostat	ICH E6(R3); 21 CFR Part 11 audit trail	Independent verification; segregation from blinded team
S3-38	**Protocol-deviation classification & reconciliation log (important vs non-important)**	Clinical ops + biostat (blinded review)	ICH E3; ICH E6(R3); ICH E9(R1) (impact on populations/estimands)	Blinded review meeting; medical/biostat adjudication documented
S3-39	**Analysis-population derivation & participant-disposition reconciliation**	Lead biostatistician + programming	CDISC ADaMIG v1.3 (ADSL population flags); ICH E9(R1)	Independent double-programming + blinded allocation sign-off

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	(population-flag QC)			
S3-40	**Data Review Meeting minutes & blinded data-review sign-off (pre-lock)**	Cross-functional (DM, biostat, clinical, PK), blinded	ICH E6(R3); ALCOA++; 21 CFR Part 11	Attendees confirm minutes; signatures retained
S3-41	**Database lock authorization & 21 CFR Part 11 audit-trail review / system records**	Data management + QA + biostat	21 CFR Part 11; ALCOA++; ICH E6(R3)	QA review of audit trail; independent lock approval
S3-42	**PK sampling-time deviation & actual-vs-nominal time reconciliation (PK timing dataset QC)**	PK scientist + biostat programming	CDISC PC/PP domains; FDA PK guidance; ALCOA++	Independent check vs source; PK scientist sign-off

Stage gate: Database lock. 42 deliverables.

S4: Analysis & QC

TLFs and PK/PD analyses are produced and independently double-programmed; reported numbers come from validated programs and engines (Phoenix WinNonlin or PKNCA for NCA).

ID	Deliverable	Owner	Standard	QC
S4-01	**Analysis environment qualification & program setup (folder structure, validated software versions, run/seed control)**	Stat programmer	21 CFR Part 11; ALCOA++; ICH E6(R3) (computerized systems / software validation); CSV/IQ-OQ-PQ	Installation/operational qualification records; environment qualification checklist; version log; reviewer sign-off
S4-02	**Disposition tables (screening, randomization, dosed, completed/discontinued; analysis-set attribution)**	Stat programmer	ICH E3; ADaMIG v1.3 (ADSL); SAP	Independent double-programming + reconciliation against shells; population counts cross-checked to ADSL
S4-03	**Demographics & baseline characteristics tables**	Stat programmer	ICH E3; ADaMIG v1.3; CDISC CT; SAP	Independent double-programming + reconciliation; descriptive stats verified against ADSL/raw
S4-04	**Exposure / dosing & treatment-compliance tables**	Stat programmer	ICH E3; SDTMIG v3.4 (EX); ADaMIG v1.3; SAP	Independent double-programming + reconciliation; dosing records cross-checked to EX/EC
S4-05	**Protocol-deviation summary tables (with importance classification)**	Stat programmer	ICH E3; ICH E6(R3); SDTMIG v3.4 (DV); SAP	Independent double-programming + reconciliation; classification reviewed at blinded data review
S4-06	**PK concentration summary tables + mean concentration-time figures (linear &**	Stat programmer	ICH E3; ADaMIG v1.3 (ADPC); SDTMIG v3.4 (PC); SAP / PK analysis plan	Independent double-programming + reconciliation; BLQ handling verified;

ID	Deliverable	Owner	Standard	QC
	semi-log)			figures QC'd vs tables
S4-07	**NCA execution in validated Phoenix WinNonlin (PK parameter derivation)**	PK scientist	Validated NCA in Phoenix WinNonlin; ICH E3; ADaMIG v1.3 (ADPP); SAP / PK analysis plan	Phoenix is the validated engine of record; independent re-derivation of key parameters (second analyst / second tool) + reconciliation; lambda_z selection reviewed; run records archived
S4-08	**PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)**	Stat programmer	ICH E3; ADaMIG v1.3 (ADPP); SAP	Independent double-programming of the summary table + reconciliation; values traceable to validated Phoenix ADPP
S4-09	**Dose-proportionality & accumulation analyses**	Stat programmer	ICH E3; SAP / PK analysis plan; relevant FDA/EMA dose-proportionality guidance	Independent double-programming + reconciliation; model specification reviewed; outputs traced to validated ADPP
S4-10	**Food-effect / DDI / relative BA-BE statistical analysis (ANOVA, 90% CI, GMR), if applicable**	Stat programmer	ICH E3; FDA/EMA BE & food-effect & DDI guidances; SAP	Independent double-programming of mixed-model/GMR/90% CI + reconciliation; model statement (fixed/random effects) reviewed; values traced to validated ADPP
S4-11	**Concentration-QTc (C-QTc) / QT-QTc analysis, if applicable**	Pharmacometrician	ICH E14 + E14 Q&A (R3) (Q5.1, concentration-response modeling) and E14/S7B Q&As (2022); ICH E3; ADaMIG v1.3 (ADEG); SAP	Independent re-modeling / double-programming + reconciliation; C-QTc model assumptions reviewed; run records archived
S4-12	**popPK / exposure-response modeling (NONMEM/Monolix) with run records**	Pharmacometrician	Validated NONMEM/Monolix; popPK analysis plan; FDA/EMA popPK & exposure-response guidance; ICH E3	Run records (control streams/seeds/logs) archived; independent review of model code & assumptions; key results reproduced; VPC/GOF reviewed
S4-13	**Safety TLFs: TEAEs (overview, by SOC/PT, severity, relationship, by dose)**	Stat programmer	ICH E3; MedDRA; ADaMIG v1.3 (ADAE); SAP	Independent double-programming + reconciliation; TEAE flag/derivation verified; counts cross-checked to ADAE
S4-14	**Safety TLFs: deaths, SAEs, and AEs leading to discontinuation**	Stat programmer	ICH E3; ICH E2A/E6(R3); MedDRA; SAP	Independent double-programming + reconciliation; SAE/death flags reconciled with safety database/PV
S4-15	**Safety TLFs: clinical laboratory analyses**	Stat programmer	ICH E3; CDISC CT; ADaMIG v1.3 (ADLB);	Independent double-programming +

ID	Deliverable	Owner	Standard	QC
	(summary, change-from-baseline, shift tables, outliers, Hy's law)		SAP	reconciliation; reference-range/grade derivations verified
S4-16	**Safety TLFs: vital signs, ECG (12-lead parameters), physical exam**	Stat programmer	ICH E3; ICH E14 (ECG categorical); ADaMIG v1.3 (ADVS/ADEG); SAP	Independent double-programming + reconciliation; baseline/CFB and outlier criteria verified
S4-17	**PD endpoint analyses (biomarker / target-engagement summaries and PK/PD)**	Pharmacometrician	ICH E3; ADaMIG v1.3 (ADPD); SAP	Independent double-programming + reconciliation; derivations/units verified; PK/PD models reviewed
S4-18	**Immunogenicity / anti-drug antibody (ADA) analyses, if applicable**	Stat programmer	ICH E3; FDA/EMA immunogenicity guidance; CDISC IS; SAP	Independent double-programming + reconciliation; TE-ADA derivation verified
S4-19	**Exploratory / post-hoc analyses**	Lead biostatistician	ICH E3; ICH E9; SAP (exploratory designation)	Risk-based QC (key exploratory outputs double-programmed); exploratory flagging reviewed
S4-20	**Statistical analysis execution (run-all production of SAP-specified analyses)**	Stat programmer	ICH E9; ICH E3; 21 CFR Part 11; ALCOA++; SAP	Log review (no errors/warnings); output-vs-shell check; independent double-programming reconciliation gate before release
S4-21	**Independent double-programming QC + reconciliation (discrepancy resolution)**	Stat programmer	ICH E6(R3); 21 CFR Part 11; ALCOA++; validation/QC plan; SAP	This IS the QC step: independent double-programming + documented reconciliation; lead biostatistician sign-off
S4-22	**TLF validation / QC documentation & discrepancy log**	Stat programmer	21 CFR Part 11; ALCOA++; ICH E6(R3); validation/QC plan	Completeness/ consistency review of QC records; lead biostatistician sign-off
S4-23	**Blinded / blind data-review meeting & documented decisions**	Lead biostatistician	ICH E9; ICH E6(R3); ICH E3; SAP / Data Review Plan	Decisions documented before unblinding; cross-functional sign-off; decisions locked
S4-24	**Unblinding / code-break execution**	Lead biostatistician	ICH E6(R3); 21 CFR Part 11 (audit trail/access); unblinding/code-break SOP	Authorization & timing documented; access-controlled execution; audit trail; segregation of duties verified
S4-25	**Pharmacometrics / popPK analysis report**	Pharmacometrician	FDA/EMA popPK & exposure-response guidance; ICH E3; modeling analysis plan	Independent review of model & report; key results reproduced; run-record completeness check
S4-26	**QC documentation / validation records package (analysis**	Lead biostatistician	21 CFR Part 11; ALCOA++; ICH E6(R3); validation/QC plan	Completeness/ traceability review; lead biostatistician

ID	Deliverable	Owner	Standard	QC
	stage)			sign-off; archived per retention policy
S4-27	**Phoenix WinNonlin NCA parameter-derivation QC / independent NCA verification record**	PK scientist + independent PK QC reviewer	Phoenix WinNonlin (validated NCA); FDA PK guidance; 21 CFR Part 11	Independent PK reviewer; documented parameter verification
S4-28	**Pharmacometrics model qualification & validation records (covariate selection, GOF, VPC, bootstrap, run log)**	Pharmacometrician + independent reviewer	NONMEM/Monolix; FDA popPK guidance; 21 CFR Part 11	Independent pharmacometrics review; reproducible run records
S4-29	**Risk-based QC plan execution & QC-level assignment matrix (per-output QC tiering)**	Statistical programming QC lead	ICH E6(R3) risk-based quality management; ICH E8(R1)	QC lead sign-off; mandatory independent double-programming for RED outputs
S4-30	**Concentration-QTc (C-QTc) model assumption-check & QC record**	Pharmacometrician / cardiac-safety statistician	ICH E14 / E14-S7B Q&A; FDA C-QTc guidance	Independent cardiac-safety statistician review

Stage gate: All outputs QC-passed. 30 deliverables.

S5: Reporting & CSR

Results become the CSR: statistical methods and results text, in-text tables/figures, appendices, the submission package, and cross-document consistency.

ID	Deliverable	Owner	Standard	QC
S5-01	**CSR Statistical Methods section (Section 9.7 / 11) per ICH E3**	Lead biostatistician	ICH E3 (Section 9.7/11 structure), ICH E9 + E9(R1) estimands, CDISC ADaMIG v1.3 (population/derivation traceability)	Peer review by second biostatistician against final SAP; cross-check that every method maps to a reported output; medical-writer editorial QC; SAP-vs-CSR methods reconciliation checklist
S5-02	**CSR Results: in-text (in-body) summary tables and figures**	Stat programmer	ICH E3 (Section 10–12 Results), CDISC ADaMIG v1.3 (source datasets), Phoenix WinNonlin (NCA engine of record)	Independent double-programming or 1:1 reconciliation of every in-text value against the corresponding full TLF and source ADaM; numerical-consistency check; QC log signed off
S5-03	**Biostatistician contribution to and review of the CSR draft (Results text, consistency QC)**	Lead biostatistician	ICH E3, ICH E9/E9(R1), 21 CFR Part 11 (review records/audit trail)	Documented number-by-number consistency check (CSR-vs-TLF-vs-ADaM); second-reviewer spot-check; tracked-changes/comment resolution log retained as a controlled record

ID	Deliverable	Owner	Standard	QC
S5-04	**CSR Appendix 16.1.9: Documentation of statistical methods incl. final SAP**	Lead biostatistician	ICH E3 (Appendix 16.1.9), ICH E9/E9(R1), 21 CFR Part 11 / ALCOA++ (versioning, audit trail)	Completeness/version reconciliation (approved versions only); cross-check SAP content vs CSR Methods; QA review of appendix assembly
S5-05	**CSR Appendix 16.2: Data listings (by participant)**	Stat programmer	ICH E3 (Section 16.2), CDISC SDTMIG v3.4 / ADaMIG v1.3	Independent double-programming or QC review of each listing; reconciliation against summary TLFs; QC log
S5-06	**CSR appendices: Define-XML + reviewer guides (cSDRG/ADRG) and annotated CRF as submission appendices**	Stat programmer	CDISC Define-XML v2.1, cSDRG/ADRG templates, CDISC Controlled Terminology, FDA/PMDA submission requirements	Pinnacle 21 conformance re-run on final datasets; reviewer-guide completeness check; define-vs-dataset consistency QC
S5-07	**Patient narrative data support (safety narratives)**	Stat programmer	ICH E3 (Section 12.3.2 / narratives), ICH E6(R3) GCP, 21 CFR Part 11, ALCOA++	Narrative-to-source data reconciliation per participant; independent check of the trigger list (all deaths/SAEs captured); QC log
S5-08	**Participant (patient) profiles**	Stat programmer	ICH E3, CDISC SDTMIG v3.4 / ADaMIG v1.3, 21 CFR Part 11	Spot-check profiles vs source listings; independent review of profile program; reconciliation against ADaM
S5-09	**Submission datasets + analysis programs as regulatory deliverables (with ADRG/cSDRG)**	Stat programmer	CDISC SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, Controlled Terminology; FDA Study Data Technical Conformance Guide (v6.3, Oct 2026); 21 CFR Part 11	Final Pinnacle 21 conformance; independent double-programming evidence retained; program executability/QC review; traceability raw→SDTM→ADaM→TLF confirmed
S5-10	**Traceability matrix (raw → SDTM → ADaM → TLF/CSR)**	Lead biostatistician	CDISC ADaMIG v1.3 (traceability), Define-XML v2.1 value-level metadata, ICH E3, 21 CFR Part 11 / ALCOA++	Independent verification that each CSR/TLF number traces to a derivation and source; gap review; QA audit
S5-11	**QC of CSR-vs-TLF-vs-dataset numerical consistency**	Lead biostatistician	ICH E3, ICH E9, 21 CFR Part 11 / ALCOA++ (QC records, audit trail)	Independent reviewer performs the check; discrepancy log tracked to closure; second-level QA sign-off
S5-12	**Statistical input to clinical study synopsis and briefing documents**	Lead biostatistician	ICH E3 (synopsis), ICH E9/E9(R1), ICH M11 alignment for protocol/briefing context	Consistency check vs CSR body and TLFs (numbers identical); peer/medical review; sign-off
S5-13	**Statistical/biometrics sign-off and CSR**	Lead biostatistician	ICH E3, ICH E6(R3) GCP, 21 CFR Part 11 /	Final QA review; confirmation all

ID	Deliverable	Owner	Standard	QC
	release for submission		ALCOA++ (e-signatures, audit trail), CDISC submission standards	discrepancy logs closed; e-signature with audit trail; engine-of-record records (software and version) attached
S5-14	**eCTD Module 5 submission package assembly & study-data standards conformance (study data reviewer's guide bundle, study-tagging)**	Regulatory + biometrics submission lead	FDA Study Data Technical Conformance Guide; eCTD; Define-XML v2.1	Submission-readiness review; Pinnacle 21 package validation
S5-15	**Analysis Data Reviewer's Guide of record finalization & cross-document consistency (SAP ↔ Define ↔ ADRG ↔ CSR)**	Lead biostatistician + metadata lead	ICH E3; ADaMIG v1.3; Define-XML v2.1; cSDRG/ADRG	Biostatistician sign-off; documented discrepancy closure

Stage gate: CSR draft delivered. 15 deliverables.

CDISC dataset & metadata inventory

The standardized-data spine that makes the study submission-ready (SDTM → ADaM → Define-XML, with conformance and reviewer guides).

ID	Deliverable	Standard	QC
S3-04	SDTM datasets: DM (Demographics)	CDISC SDTMIG v3.4, Controlled Terminology, Define-XML v2.1	Independent double-programming + Pinnacle 21 conformance; reconciliation to raw and randomization
S3-05	SDTM datasets: PC (Pharmacokinetic Concentrations)	CDISC SDTMIG v3.4 (Findings), Define-XML v2.1, value-level metadata; CT for units/analytes	Independent double-programming + Pinnacle 21; 100% reconciliation to bioanalytical transfer (record count, value, BLQ flags); nominal-vs-actual time check
S3-06	SDTM datasets: PP (Pharmacokinetic Parameters)	CDISC SDTMIG v3.4 (PP domain), CT for PK parameters (PKPARMCD), Define-XML v2.1, RELREC linkage to PC	Reconciliation to Phoenix output (the engine of record); independent double-programming of the SDTM structure; Pinnacle 21
S3-07	SDTM datasets: EX (Exposure)	CDISC SDTMIG v3.4 (Interventions), Define-XML v2.1, Controlled Terminology	Independent double-programming + Pinnacle 21; reconciliation to randomization and to PC time-after-dose derivation
S3-08	SDTM datasets: AE (Adverse Events)	CDISC SDTMIG v3.4 (Events), MedDRA (versioned), Define-XML v2.1, CT	Independent double-programming + Pinnacle 21; MedDRA coding QC; reconciliation to safety database/SAE recon
S3-09	SDTM datasets: LB (Laboratory)	CDISC SDTMIG v3.4 (Findings), CT/LOINC, Define-XML v2.1 value-level metadata, unit	Independent double-programming + Pinnacle 21; 100% reconciliation to lab

ID	Deliverable	Standard	QC
		standardization	transfer; unit conversion QC
S3-10	SDTM datasets: VS (Vital Signs)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to raw
S3-11	SDTM datasets: EG (ECG)	CDISC SDTMIG v3.4 (Findings), CT, Define-XML v2.1; time-matched to PC for C-QTc	Independent double-programming + Pinnacle 21; reconciliation to ECG core-lab transfer; PK time-matching QC
S3-12	SDTM datasets: DS (Disposition)	CDISC SDTMIG v3.4 (Events), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; reconciliation to enrollment/randomization counts
S3-13	SDTM datasets: MH (Medical History)	CDISC SDTMIG v3.4 (Events), MedDRA, CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-14	SDTM datasets: CM (Concomitant/Prior Medications)	CDISC SDTMIG v3.4 (Interventions), WHODrug (versioned), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; coding QC
S3-15	SDTM datasets: PD/DV and other supporting domains (PD/PE/SU/QS, DV, CO, SE/TA/TV/TE trial design, SUPPQUAL, RELREC)	CDISC SDTMIG v3.4 (Findings/Events/Special-Purpose/Trial-Design/SUPP/RELREC), CT, Define-XML v2.1	Independent double-programming + Pinnacle 21; RELREC integrity check (PC<->PP); deviation-classification reconciliation
S3-16	SDTM Define-XML (define.xml + stylesheet)	CDISC Define-XML v2.1, SDTMIG v3.4, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; peer review; origin/aCRF cross-link check
S3-17	SDTM Reviewer Guide (cSDRG)	PHUSE cSDRG template, CDISC SDTMIG v3.4, Define-XML v2.1	Peer review; consistency with Define-XML and Pinnacle 21 explanations; sign-off
S3-18	ADaM datasets: ADSL (participant-level analysis dataset)	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21 ADaM checks; population-flag reconciliation to disposition/randomization
S3-19	ADaM datasets: ADPC (PK Concentrations Analysis)	CDISC ADaMIG v1.3 (BDS / PK ADaM), Define-XML v2.1, traceability to PC/EX, CT	Independent double-programming + Pinnacle 21; time-after-dose & BLQ-rule QC; reconciliation to PC
S3-20	ADaM datasets: ADPP (PK Parameters Analysis)	CDISC ADaMIG v1.3, Define-XML v2.1, traceability to PP/Phoenix, CT (PK params)	Reconciliation to Phoenix (engine of record); independent double-programming; Pinnacle 21
S3-21	ADaM datasets: ADAE (Adverse Events Analysis)	CDISC ADaMIG v1.3 OCCDS, MedDRA, Define-XML v2.1, traceability to AE, CT	Independent double-programming + Pinnacle 21; TRTEMFL logic QC; reconciliation to SDTM AE & SAE database
S3-22	ADaM datasets: ADLB (Laboratory Analysis)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1 value-level metadata, traceability to LB, CT	Independent double-programming + Pinnacle 21; baseline/change & category-derivation QC; reconciliation to LB
S3-23	ADaM datasets: ADVS (Vital	CDISC ADaMIG v1.3 BDS,	Independent double-

ID	Deliverable	Standard	QC
	Signs Analysis)	Define-XML v2.1, traceability to VS, CT	programming + Pinnacle 21; baseline/change QC; reconciliation to VS
S3-24	ADaM datasets: ADEG (ECG Analysis)	CDISC ADaMIG v1.3 BDS, Define-XML v2.1, traceability to EG/PC, CT; ICH E14/concentration-QTc practice	Independent double-programming + Pinnacle 21; QTc category & time-matching QC; reconciliation to EG
S3-25	ADaM datasets: ADPD and other analysis datasets (ADPD, ADCM, ADMH, deviation/exposure analysis as applicable)	CDISC ADaMIG v1.3 BDS/OCCDS, Define-XML v2.1, traceability to SDTM, CT	Independent double-programming + Pinnacle 21; PK-PD linkage & baseline/change QC
S3-26	ADaM Define-XML (define.xml + stylesheet)	CDISC Define-XML v2.1, ADaMIG v1.3, Controlled Terminology	Pinnacle 21 Define conformance + dataset-define consistency; derivation-method review; traceability check
S3-27	ADaM Reviewer Guide (ADRG)	PHUSE ADRG template, CDISC ADaMIG v1.3, Define-XML v2.1	Peer review; consistency with Define-XML, SAP and Pinnacle 21 explanations; sign-off
S3-28	CDISC conformance / validation (Pinnacle 21) report: SDTM & ADaM	CDISC conformance rules, SDTMIG v3.4, ADaMIG v1.3, Define-XML v2.1, FDA/PMDA validator rules	Validated Pinnacle 21 engine; documented triage/disposition of every finding; reviewer-guide reconciliation
S3-31	Dry-run / test datasets & dry-run TLF execution	Sponsor validation/QC plan; CDISC SDTMIG v3.4/ADaMIG v1.3; 21 CFR Part 11 (program version control)	Independent double-programming on dry-run; shell-vs-output reconciliation; defect tracking to closure
S3-34	Traceability (raw -> SDTM -> ADaM) matrix	CDISC SDTMIG v3.4/ADaMIG v1.3/Define-XML v2.1 traceability principles; ICH E6(R3), ALCOA++	Cross-link verification raw<->SDTM<->ADaM; peer review; consistency with define/reviewer guides
S3-35	Controlled Terminology & coding-dictionary version management (CDISC CT, MedDRA, WHODrug)	CDISC Controlled Terminology (versioned), MedDRA, WHODrug; Define-XML v2.1; 21 CFR Part 11	Version-lock verification; CT-conformance via Pinnacle 21; coding QC against locked dictionaries
S3-42	PK sampling-time deviation & actual-vs-nominal time reconciliation (PK timing dataset QC)	CDISC PC/PP domains; FDA PK guidance; ALCOA++	Independent check vs source; PK scientist sign-off

TLF & analysis-output inventory (Stage 4)

Tables, listings, and figures by domain, plus the PK/PD analyses. Reported numbers come from validated programs and engines and are independently double-programmed.

ID	Deliverable	Owner	QC
S4-02	Disposition tables (screening, randomization, dosed, completed/discontinued; analysis-set attribution)	Stat programmer	Independent double-programming + reconciliation against shells; population counts cross-checked to ADSL
S4-03	Demographics & baseline characteristics tables	Stat programmer	Independent double-programming + reconciliation;

ID	Deliverable	Owner	QC
			descriptive stats verified against ADSL/raw
S4-04	Exposure / dosing & treatment-compliance tables	Stat programmer	Independent double-programming + reconciliation; dosing records cross-checked to EX/EC
S4-05	Protocol-deviation summary tables (with importance classification)	Stat programmer	Independent double-programming + reconciliation; classification reviewed at blinded data review
S4-06	PK concentration summary tables + mean concentration-time figures (linear & semi-log)	Stat programmer	Independent double-programming + reconciliation; BLQ handling verified; figures QC'd vs tables
S4-07	NCA execution in validated Phoenix WinNonlin (PK parameter derivation)	PK scientist	Phoenix is the validated engine of record; independent re-derivation of key parameters (second analyst / second tool) + reconciliation; lambda_z selection reviewed; run records archived
S4-08	PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)	Stat programmer	Independent double-programming of the summary table + reconciliation; values traceable to validated Phoenix ADPP
S4-09	Dose-proportionality & accumulation analyses	Stat programmer	Independent double-programming + reconciliation; model specification reviewed; outputs traced to validated ADPP
S4-11	Concentration-QTc (C-QTc) / QT-QTc analysis, if applicable	Pharmacometrician	Independent re-modeling / double-programming + reconciliation; C-QTc model assumptions reviewed; run records archived
S4-12	popPK / exposure-response modeling (NONMEM/Monolix) with run records	Pharmacometrician	Run records (control streams/seeds/logs) archived; independent review of model code & assumptions; key results reproduced; VPC/GOF reviewed
S4-13	Safety TLFs: TEAEs (overview, by SOC/PT, severity, relationship, by dose)	Stat programmer	Independent double-programming + reconciliation; TEAE flag/derivation verified; counts cross-checked to ADAE
S4-14	Safety TLFs: deaths, SAEs, and AEs leading to discontinuation	Stat programmer	Independent double-programming + reconciliation; SAE/death flags reconciled with safety database/PV
S4-15	Safety TLFs: clinical laboratory analyses (summary, change-from-baseline, shift tables, outliers, Hy's law)	Stat programmer	Independent double-programming + reconciliation; reference-range/grade derivations verified
S4-16	Safety TLFs: vital signs, ECG (12-lead parameters), physical exam	Stat programmer	Independent double-programming + reconciliation; baseline/CFB and outlier

ID	Deliverable	Owner	QC
			criteria verified
S4-18	Immunogenicity / anti-drug antibody (ADA) analyses, if applicable	Stat programmer	Independent double-programming + reconciliation; TE-ADA derivation verified
S4-22	TLF validation / QC documentation & discrepancy log	Stat programmer	Completeness/consistency review of QC records; lead biostatistician sign-off
S4-25	Pharmacometrics / popPK analysis report	Pharmacometrician	Independent review of model & report; key results reproduced; run-record completeness check
S4-27	Phoenix WinNonlin NCA parameter-derivation QC / independent NCA verification record	PK scientist + independent PK QC reviewer	Independent PK reviewer; documented parameter verification
S4-28	Pharmacometrics model qualification & validation records (covariate selection, GOF, VPC, bootstrap, run log)	Pharmacometrician + independent reviewer	Independent pharmacometrics review; reproducible run records
S4-30	Concentration-QTc (C-QTc) model assumption-check & QC record	Pharmacometrician / cardiac-safety statistician	Independent cardiac-safety statistician review

Validation & quality (cross-cutting)

- **Independent double programming** of every submission-grade output, with every discrepancy logged and resolved.
- **Engine of record** for every deliverable: reported numbers come from validated, double-programmed SAS/R programs on ADaM; NCA from Phoenix WinNonlin (version per the ADRG, currently 8.8) or PKNCA; CDISC conformance is checked with Pinnacle 21 (FDA engine 2304.3). Software version and build are documented, as FDA's statistical software statement asks.
- **Data integrity:** records that are attributable, legible, contemporaneous, original, accurate, complete, secure, and reliable (ICH E6(R3)); audit trails and e-signatures in systems designed to meet 21 CFR Part 11.
- **Software / environment qualification** (IQ/OQ/PQ) for SAS/R, Phoenix WinNonlin, NONMEM/Monolix, and Pinnacle 21.
- **Named sign-off:** a qualified statistician reviews and signs each deliverable.

Appendix: AI-use proposal

A proposal, not an adopted policy, for where AI tools could assist on these deliverables and where they may not. For context, FDA's guidance on using AI to support regulatory decision-making for drugs is still a draft (January 2025), and the FDA-EMA guiding principles for AI in drug development (January 2026) are non-binding.

1. Every reported number comes from a validated program or engine. An AI tool may draft code that is one side of independent double programming, never both sides.
2. Participant-level, unblinded, or identifiable data stays inside the validated environment (a local, version-pinned model, if any model is used). Only de-identified text and specifications may go to a contracted cloud service under a data-processing agreement, never a consumer account.
3. A named statistician reviews and signs every output, and AI use is recorded.

Tier	Deliverables	What an AI tool may do
Green	44	Draft text, scaffolds, and documentation from de-identified content
Amber	54	Draft dataset or TLF code as one side of double programming
Red	34	Nothing: validated programs and engines produce the numbers, and a human checks them

If AI tools are used, four records are kept:

- **AI-01 AI-use plan: data classification and tool routing** (Stage 2, before any AI-assisted work): Per planned deliverable: the data-sensitivity class and where any AI assistance may run. Participant-level, unblinded or identifiable data: a local, version-pinned model with no data leaving the validated environment. De-identified text and specifications: a contracted cloud service under a data-processing agreement, never a consumer account. Reported numbers: validated programs and engines only.
- **AI-02 AI-use log (data stage)** (Stage 3, kept current): Per Stage 3 deliverable: the data class, where AI assistance ran, the engine of record for any dataset, and the model version used.
- **AI-03 AI-use log (analysis stage)** (Stage 4, per output): Per output: the data class, where AI assistance ran, the engine of record for reported numbers (for example Phoenix WinNonlin, PKNCA or NONMEM), the model version, and retained prompts and outputs for AI-drafted code that was one side of double programming.
- **AI-04 AI-use summary for the study file** (Stage 5, before release): One summary across the study: model versions, the engine-of-record register showing every reported number came from a validated program or engine, and the retained prompt and output records.

The tier and AI note for each deliverable are on the spreadsheet's "AI use (proposal)" sheet.